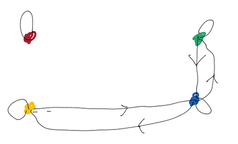
1. Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
b) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
c) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
d) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

0 1 2 3

a) yes, trivial

d)



yes

reflexive, transitive, symmetric

e) Not reflexive for vertice 2, thus not an equiv. Rel.

11. Show that the relation R consisting of all pairs (x, y) such that x and y are bit strings of length three or more that agree in their first three bits is an equivalence relation on the set of all bit strings of length three or more.

for an element that is identical, the first three bits will match. R is thus reflexive.

equality is a commutative operation, thus R is symmetric

as equality is a transitive operation R is also transitive

$\therefore$ R is an equil. Rel.

30. What are the equivalence classes of these bit strings for the equivalence relation in Exercise 11?

a) 010 **b)** 1011 **c)** 11111 **d)** 01010101

a) $\{x \mid \text{first 3 bits} = 010\}$

b) $\{x \mid \text{first 3 bits} = 101\}$

c) $\{x \mid \text{first 3 bits} = 111\}$

d) $\{x \mid \text{first 3 bits} = 010\}$

. Draw graph models, stating the type of graph (from Table 1) used, to represent airline routes where every day there are four flights from Boston to Newark, two flights from Newark to Boston, three flights from Newark to Miami, two flights from Miami to Newark, one flight from Newark to Detroit, two flights from Detroit to Newark, three flights from Newark to Washington, two flights from Washington to Newark, and one flight from Washington to Miami, with

a) an edge between vertices representing cities that have a flight between them (in either direction).

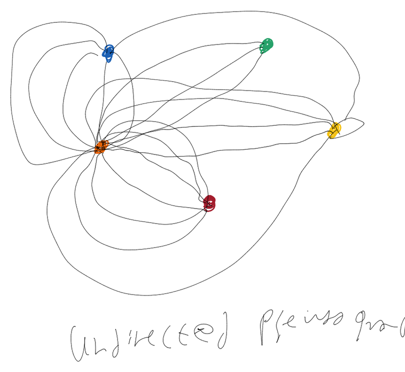
b) an edge between vertices representing cities for each flight that operates between them (in either direction).

c) an edge between vertices representing cities for each flight that operates between them (in either direction), plus a loop for a special sightseeing trip that takes off and lands in Miami.

10. For each undirected graph in Exercises 3–9 that is simple, find a set of edges to remove to make it simp

11. Let G be a simple graph. Show that the relation R on set of vertices of G such that uRv if and only if ther

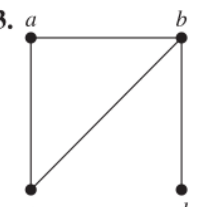
Boston Newark Miami Detroit Washington



undirected pseudo graph

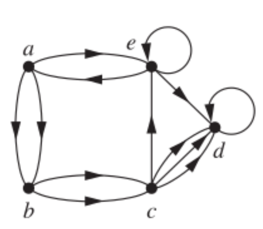
For Exercises 3–9, determine whether the graph shown has directed or undirected edges, whether it has multiple edges, and whether it has one or more loops. Use your answers to determine the type of graph in Table 1 this graph is.

3.



undirected simple

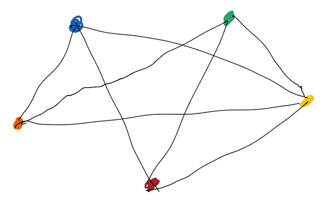
8.



Directed pseudo

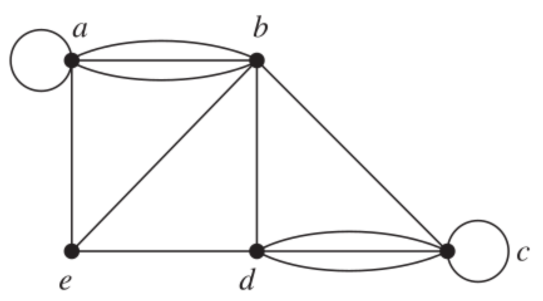
13. The **intersection graph** of a collection of sets $A_1, A_2, \ldots, A_n$ is the graph that has a vertex for each of these sets and has an edge connecting the vertices representing two sets if these sets have a nonempty intersection. Construct the intersection graph of these collections of sets.

a) $A_1 = \{0, 2, 4, 6, 8\}$, $A_2 = \{0, 1, 2, 3, 4\}$, $A_3 = \{1, 3, 5, 7, 9\}$, $A_4 = \{5, 6, 7, 8, 9\}$, $A_5 = \{0, 1, 8, 9\}$



In Exercises 1–3 find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.

2.



#V=5
#E=13

degrees
a: 6
b: 6
c: 6
d: 5
e: 3

we didn't talk about isolated/pendant

5. Can a simple graph exist with 15 vertices each of degree five?

by the handshaking Lemma,

$\sum \deg(v) = 2|E|$

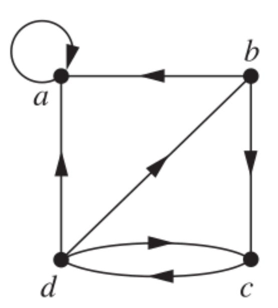
$15 \cdot 5 = 75$

$\frac{75}{2} \notin \mathbb{Z}$

$|E| \notin \mathbb{Z}$, thus a simple graph w/ $|V| = 15$ and each $\deg(v) = 5$ cannot exist

In Exercises 7–9 determine the number of vertices and edges and find the in-degree and out-degree of each vertex for the given directed multigraph.

7.



#V=4
#E=7

Degrees in
a: 3
b: 1
c: 2
d: 1

Degrees out
a: 1
b: 2
c: 1
d: 3

10. For each of the graphs in Exercises 7–9 determine the sum of the in-degrees of the vertices and the sum of the out-degrees of the vertices directly. Show that they are both equal to the number of edges in the graph.

in: 3+1+2+1=7 out: 1+2+1+3=7

#V=7